

BLK II - UNIT 5: METOC Mission Products

f. METOC Mission Briefs

Flight Weather Brief Format

Part 1: Local Surface Data

PART I - LOCAL SURFACE DATA							
1. DATE	2. Mission Name	3. DEP PT/ETD	4. RWY TEMP	5. DEWPOINT	6. ALTIMETER	7. VIS	8. MISSION CONDITION
		Z	°F/C	°F/C		SM	
9. SFC WIND	10. CLIMB WINDS SFC, FL050, FL100		11. LOCAL WEATHER WATCH/WARNING/ADVISORY				
13. REMARKS							

Box 1 – Date: The date of the mission

Box 2 – Mission Name: The name given to this mission

Box 3 – Departure Point / Estimated Time of Departure: The station ICAO where you will begin the mission, and the time the mission is expected to begin in Zulu time.

Box 4/5 – Runway Temperature / Dewpoint: The forecasted temperature and dewpoint for the time listed in box 3. Give both Fahrenheit and Celsius temperatures.

Box 6 – Altimeter: The forecasted altimeter value in INS for the time listed in box 3.

Box 7 – Visibility: The forecasted visibility in SM for the time listed in box 3.

Box 8 – Mission Condition: Based on the mission thresholds, input the mission condition for the start of the mission in this box. If you input marginal or unfavorable, this would require justification in box 13.

Box 9 – Surface Winds: Forecasted surface winds for the time listed in box 3.

Box 10 – Climb Winds: For aviation missions, input the wind data for surface winds, 5,000 ft winds, and 10,000 ft winds.

Box 11 – Local WWAs: If there are any active WWAs that will still be valid for the time listed in box 3, or if you are forecasting conditions that could potentially lead to issuing a WWA for that time, input the WWA in this box.

Box 13 – Remarks: If there is any other data you feel is important or that you feel the mission planner should know, input that information in plain language in this box. This box will also house your justification for any marginal or unfavorable mission conditions.

Part 2: Enroute & Mission Data

PART II - ENROUTE & MISSION DATA																	
14. FLT LEVEL/WINDS/TEMP					15. SPACE WEATHER					16. SOLAR/LUNAR							
					NO IMPACT MARGINAL SEVERE					BMNT		Z					
					FREQ					SR		Z		MR		Z	
					GPS					SS		Z		MS		Z	
					RAD					EENT		Z		ILLUM		%	
17. CLOUDS AT FLT LEVEL					18. OBSCURATIONS RESTRICTING VISIBILITY												
☐ YES ☐ NO ☐ IN AND OUT					☐ YES ☐ NO TYPE												
19. MINIMUM CEILING - LOCATION					20. MAXIMUM CLOUD TOPS - LOCATION					21. MINIMUM FREEZING LVL - LOCATION							
FT AGL					FT MSL					FT MSL							
22. THUNDERSTORMS				23. TURBULENCE				24. ICING				25. PRECIPITATION					
CHART				CHART				CHART				CHART					
☐ NONE ☐ AREA ☐ LINE				☐ NONE ☐ IN CLEAR ☐ IN CLOUD				☐ NONE ☐ RIME ☐ MIXED ☐ CLEAR				☐ NONE ☐ RAIN ☐ SNOW					
ISOLATED 1 - 2%				LIGHT				TRACE				LIGHT					
FEW 3 - 15%				MODERATE				LIGHT				MODERATE					
SCATTERED 16 - 45%				SEVERE				MODERATE				HEAVY					
NUMEROUS - MORE THAN 45%				EXTREME				SEVERE				SHOWERS					
HAIL, SEVERE TURBULENCE & ICING, HEAVY PRECIPITATION, LIGHTNING & WIND SHEAR EXPECTED IN AND NEAR THUNDERSTORMS.				LEVELS				LEVELS				FREEZING					
				LOCATION				LOCATION				LOCATION					

Box 14 – Flight Level Winds / Temperature: Based on the given flight level for your mission, input the temperature and wind data in this section for what the pilot can expect at their flight level.

Box 15 – Space Weather: Navigate to the space weather bulletin and input the date and time window of the space impacts, the severity of the impact and the type.

Box 16 – Solar / Lunar: Navigate to the solar/lunar data on Bifrost and access the ICAO closest to your mission location. Ensure you gather data for the correct date and input the solar / lunar data onto your brief in Zulu time. If there is no ICAO located near your mission location, navigate to the US Naval Observatory Solar / Lunar data and use the latitude / longitude of your mission location to gather this data. This data is useful for every mission type to determine the best course of action if any amount of stealth is required.

Box 17 – Clouds at Flight Level: Forecast the amount of cloud coverage the pilot can expect at their flight level.

Box 18 – Obscurations Restricting Visibility: At the pilot's flight level, input if there is any expected obscurations they should be aware of. If you have already indicated the presence of clouds at flight level, you do not need to put that in this section. This box is for any other phenomena that could restrict visibility.

Box 19 – Minimum Ceiling Location: Along the route of the mission, input the lowest ceiling if applicable and general location of them. If no clouds are forecast, you can put N/A in this section.

Box 20 – Maximum Cloud Tops: Along the route of the mission, input the highest cloud tops expected and the general location of those tops. If no clouds are forecast, you can put N/A in this section.

Box 21 – Minimum Freezing Level: Along the route of the mission, input the lowest freezing level height expected and the general location of the low freezing level.

Box 22 – Thunderstorms: First, input the timeframe of the hazard chart you used to gather thunderstorm data. Next, indicate whether there are no thunderstorms along the mission route, if the thunderstorms are covering an area, or if the thunderstorms are in a line. Then mark the appropriate amount of coverage based on the hazard charts and include the general location of the thunderstorms.

Box 23 – Turbulence: First, input the timeframe of the hazard chart you used to gather turbulence data. Next, indicate the type, severity, layers, and location of forecasted turbulence along the mission route.

Box 24 – Icing: First, input the timeframe of the hazard chart you used to gather icing data. Next, indicate the type, severity, layers, and location of forecasted icing along the mission route.

Box 25 – Precipitation: If there is any forecasted precipitation along your mission route, input the time frame in the first box, then indicate the type as either rain or snow, then mark the intensity of the forecasted precipitation. Last, add the general location precipitation is expected.

Part 3: Mission Execution Forecast

PART III - Mission Execution Forecast								
26.	27. VALID TIME		28. SFC WIND	29. VSBY/WEA	30. CLOUD LAYERS	31. ALTIMETER	32. MAX TEMP	33. MIN TEMP
DESTINATION	Z	TO	Z			INS	F/C	F/C
ALTN DEST 1	Z	TO	Z			INS	F/C	F/C
ALTN DEST 2	Z	TO	Z			INS	F/C	F/C

Box 26 – Destination / Alternate Destinations: These boxes will contain the location / ICAO of the target area for your mission. If your mission requires multiple locations, use the alternate destination boxes to give mission planners that information.

Box 27 – Valid Time: Input the valid times for the mission in these boxes in Zulu time. These are the times that the mission is expected to operate at its target location.

Box 28 – Surface Winds: These are the winds that are expected at your target location during the valid times. If the valid time covers a significant period, input either the average wind data if nothing is expected to impact operations or put the worst wind conditions expected during the mission window.

Box 29 – Visibility / Weather: Input the forecasted surface visibility and any weather expected at your target location within the valid time. If no weather is expected, input NSW.

Box 30 – Cloud Layers: For aviation missions, input the cloud layers in this box without using the summation principle. This should alert the pilots to the total picture of cloud coverage above the target location.

Box 31 – Altimeter: Input the lowest altimeter expected for your target location during the mission valid time.

Box 32 – Maximum Temperature: Input the maximum expected temperature is Fahrenheit and Celsius for the target location during the valid times.

Box 33 – Minimum Temperature: Input the minimum expected temperature is Fahrenheit and Celsius for the target location during the valid times.

Part 4 & 5: Comments / Briefing Record

PART IV - COMMENTS/REMARKS		
35. REMARKS		
PART V - BRIEFING RECORD		
36. WX BRIEFED TIME	37. BRIEFERS INITIALS	38. PERSON RECEIVING BRIEF

Box 35 – Remarks: If there is any information you feel the mission planners need to be aware of that was not included in the brief, input that information here.

Box 36 – Weather Briefed Time: The date and time in Zulu you delivered the brief.

Box 37 – Briefer Initials: Input your initials in this block.

Box 38 – Person Receiving Brief Initials: Input either the rank and name of the individual you brief or their initials if that is the information they give you.

Ground Brief Format

Part 1: Local Surface Data

PART I - LOCAL SURFACE DATA							
1. DATE	2. Mission Name	3. DEP PT/ETD	4. TEMP	5. DEWPOINT	6. ALTIMETER	7. VIS	8. MISSION CONDITION
		Z	°F/C	°F/C		sm	
9. SFC WIND	10. PRESENT WEATHER		11. LOCAL WEATHER WATCH / WARNING / ADVISORY				
13. REMARKS							

Box 1 – Date: The date of the mission

Box 2 – Mission Name: The name given to this mission

Box 3 – Departure Point / Estimated Time of Departure: The station ICAO where you will begin the mission, and the time the mission is expected to begin in Zulu time.

Box 4/5 – Temperature / Dewpoint: The forecasted temperature and dewpoint for the time listed in box 3. Give both Fahrenheit and Celsius temperatures.

Box 6 – Altimeter: The forecasted altimeter value in INS for the time listed in box 3.

Box 7 – Visibility: The forecasted visibility in SM for the time listed in box 3.

Box 8 – Mission Condition: Based on the mission thresholds, input the mission condition for the start of the mission in this box. If you input marginal or unfavorable, this would require justification in box 13.

Box 9 – Surface Winds: Forecasted surface winds for the time listed in box 3.

Box 10 – Present Weather: Any forecasted weather elements for the time listed in box 3.

Box 11 – Local WWAs: If there are any active WWAs that will still be valid for the time listed in box 3, or if you are forecasting conditions that could potentially lead to issuing a WWA for that time, input the WWA in this box.

Box 13 – Remarks: If there is any other data you feel is important or that you feel the mission planner should know, input that information in plain language in this box. This box will also house your justification for any marginal or unfavorable mission conditions.

Part 2: Enroute & Mission Data

PART II - ENROUTE & MISSION DATA												
				15. SPACE WEATHER			16. SOLAR/ LUNAR					
					NO IMPACT	MARGINAL	SEVERE	BMNT		Z		
				FREQ				SR		Z	MR	Z
				GPS				SS		Z	MS	Z
				RAD				EENT		Z	ILLUM	%
17. CLOUDS				18. OBSCURATIONS RESTRICTING VISIBILITY								
☐ Yes ☐ No				☐ YES ☐ NO TYPE								
19. MINIMUM CEILING				20. SKY CONDITION			21. ADDITIONAL REMARKS					
22. THUNDERSTORMS		23. SFC WINDS		24. VISIBILITY		25. RELATIVE SOIL MOISTURE						
v		KT		SM		v						
☐ NONE ☐ WITHIN 5 SM ☐ WITHIN 10 SM ☐ WITHIN 25 SM ☐ WITHIN 50 SM		v		v		☐ <40% ☐ 40-70% ☐ > 70%						
LOCATION						ACTUAL AMOUNT						
						LOCATION						

Box 15 – Space Weather: Navigate to the space weather bulletin and input the date and time window of the space impacts, the severity of the impact and the type.

Box 16 – Solar / Lunar: Navigate to the solar/lunar data on Bifrost and access the ICAO closest to your mission location. Ensure you gather data for the correct date and input the solar / lunar data onto your brief in Zulu time. If there is no ICAO located near your mission location, navigate to the US Naval Observatory Solar / Lunar data and use the latitude / longitude of your mission location to gather this data. This data is useful for every mission type to determine the best course of action if any amount of stealth is required.

Box 17 – Clouds: Are there expected clouds enroute during this mission.

Box 18 – Obscurations Restricting Visibility: Input if there is any expected obscurations they should be aware of while enroute.

Box 19 – Minimum Ceiling Location: Along the route of the mission, input the lowest ceiling if applicable and general location of them. If no clouds are forecast, you can put N/A in this section.

Box 20 – Sky Condition: Along the route of the mission, input the average sky condition for the mission enroute.

Box 21 – Additional Remarks: Along the route of the mission, if there is any other information you feel the mission planner should be aware of that was not included anywhere else in this section.

Box 22 – Thunderstorms: Input the distance (if any) thunderstorms are expected while enroute. Use the dropdown to indicate the level of mission impact thunderstorms are having on your mission based on the mission threshold chart.

Box 23 – Surface Winds: Indicate the average surface winds along your mission route if the condition is favorable. If the condition is marginal or unfavorable, put the worst expected enroute wind data.

Box 24 – Visibility: Indicate the average visibility along your mission route if the condition is favorable. If the condition is marginal or unfavorable, put the worst expected enroute visibility data.

Box 25 – Relative Soil Moisture: Input the expected RSM in this section. Identify the favorability of the value, the value range, the specific percentage forecasted, and the location of the value. Ensure to identify if there are multiple values within your area of interest.

Part 3: Mission Execution Forecast

PART III - Mission Execution Forecast								
26.	27. VALID TIME		28. SFC WIND	29. VSBY/WEA	30. CLOUD LAYERS	31. ALTIMETER	32. MAX TEMP	33. MIN TEMP
DESTINATION	Z	TO	Z			INS	F/C	F/C
ALTN DEST 1	Z	TO	Z			INS	F/C	F/C
ALTN DEST 2	Z	TO	Z			INS	F/C	F/C

Box 26 – Destination / Alternate Destinations: These boxes will contain the location / ICAO of the target area for your mission. If your mission requires multiple locations, use the alternate destination boxes to give mission planners that information.

Box 27 – Valid Time: Input the valid times for the mission in these boxes in Zulu time. These are the times that the mission is expected to operate at its target location.

Box 28 – Surface Winds: These are the winds that are expected at your target location during the valid times. If the valid time covers a significant period, input either the average wind data if nothing is expected to impact operations or put the worst wind conditions expected during the mission window.

Box 29 – Visibility / Weather: Input the forecasted surface visibility and any weather expected at your target location within the valid time. If no weather is expected, input NSW.

Box 30 – Sky Condition: Input the forecasted sky condition for the mission valid times.

Box 31 – Altimeter: Input the lowest altimeter expected for your target location during the mission valid time.

Box 32 – Maximum Temperature: Input the maximum expected temperature is Fahrenheit and Celsius for the target location during the valid times.

Box 33 – Minimum Temperature: Input the minimum expected temperature is Fahrenheit and Celsius for the target location during the valid times.

Part 4 & 5: Comments / Briefing Record

PART IV - COMMENTS/REMARKS		
35. REMARKS		
PART V - BRIEFING RECORD		
36. WX BRIEFED TIME	37. BRIEFERS INITIALS	38. PERSON RECEIVING BRIEF

Box 35 – Remarks: If there is any information you feel the mission planners need to be aware of that was not included in the brief, input that information here.

Box 36 – Weather Briefed Time: The date and time in Zulu you delivered the brief.

Box 37 – Briefer Initials: Input your initials in this block.

Box 38 – Person Receiving Brief Initials: Input either the rank and name of the individual you brief or their initials if that is the information they give you.

Oceanic and Coastal Brief Format

Part 1: Local Surface Data

PART I - LOCAL SURFACE DATA							
1. DATE	2. Mission Name	3. DEP PT/ETD	4. TEMP	5. DEWPOINT	6. ALTIMETER	7. VIS	8. MISSION CONDITION
		Z	°F/C	°F/C		sm	
9. SFC WIND	10. PRESENT WEATHER		11. LOCAL WEATHER WATCH / WARNING / ADVISORY				
13. REMARKS							

Box 1 – Date: The date of the mission

Box 2 – Mission Name: The name given to this mission

Box 3 – Departure Point / Estimated Time of Departure: The station ICAO where you will begin the mission, and the time the mission is expected to begin in Zulu time.

Box 4/5 – Temperature / Dewpoint: The forecasted temperature and dewpoint for the time listed in box 3. Give both Fahrenheit and Celsius temperatures.

Box 6 – Altimeter: The forecasted altimeter value in INS for the time listed in box 3.

Box 7 – Visibility: The forecasted visibility in SM for the time listed in box 3.

Box 8 – Mission Condition: Based on the mission thresholds, input the mission condition for the start of the mission in this box. If you input marginal or unfavorable, this would require justification in box 13.

Box 9 – Surface Winds: Forecasted surface winds for the time listed in box 3.

Box 10 – Present Weather: Any forecasted weather elements for the time listed in box 3.

Box 11 – Local WWAs: If there are any active WWAs that will still be valid for the time listed in box 3, or if you are forecasting conditions that could potentially lead to issuing a WWA for that time, input the WWA in this box.

Box 13 – Remarks: If there is any other data you feel is important or that you feel the mission planner should know, input that information in plain language in this box. This box will also house your justification for any marginal or unfavorable mission conditions.

Part 2: Enroute & Mission Data

PART II - ENROUTE & MISSION DATA												
				15. SPACE WEATHER			16. SOLAR/ LUNAR					
					NO IMPACT	MARGINAL	SEVERE	BMNT		Z		
				FREQ				SR		Z	MR	Z
				GPS				SS		Z	MS	Z
				RAD				EENT		Z	ILLUM	%
17. CLOUDS				18. OBSCURATIONS RESTRICTING VISIBILITY								
☐ Yes ☐ No				☐ YES ☐ NO TYPE								
19. MINIMUM CEILING				20. SKY CONDITION			21. ADDITIONAL REMARKS					
22. THUNDERSTORMS		23. SEA STATE		24. VISIBILITY		25. PRECIPITATION						
NONE WITHIN 5 SM WITHIN 10 SM WITHIN 25 SM WITHIN 50 SM LOCATION 		SEA STATE WAVE HEIGHT FT SEA SFC TEMP F 		SM 		NONE ☐ RAIN ☐ SNOW LIGHT MODERATE HEAVY LOCATION 						

Box 15 – Space Weather: Navigate to the space weather bulletin and input the date and time window of the space impacts, the severity of the impact and the type.

Box 16 – Solar / Lunar: Navigate to the solar/lunar data on Bifrost and access the ICAO closest to your mission location. Ensure you gather data for the correct date and input the solar / lunar data onto your brief in Zulu time. If there is no ICAO located near your mission location, navigate to the US Naval Observatory Solar / Lunar data and use the latitude / longitude of your mission location to gather this data. This data is useful for every mission type to determine the best course of action if any amount of stealth is required.

Box 17 – Clouds: Are there expected clouds enroute during this mission.

Box 18 – Obscurations Restricting Visibility: Input if there is any expected obscurations they should be aware of while enroute.

Box 19 – Minimum Ceiling Location: Along the route of the mission, input the lowest ceiling if applicable and general location of them. If no clouds are forecast, you can put N/A in this section.

Box 20 – Sky Condition: Along the route of the mission, input the average sky condition for the mission enroute.

Box 21 – Additional Remarks: Along the route of the mission, if there is any other information you feel the mission planner should be aware of that was not included anywhere else in this section.

Box 22 – Thunderstorms: Input the distance (if any) thunderstorms are expected while enroute. Use the dropdown to indicate the level of mission impact thunderstorms are having on your mission based on the mission threshold chart.

Box 23 – Sea State: Indicate the forecasted sea state while enroute. Next indicate the highest wave height in feet expected and the average sea surface temperature of the mission area. Use the dropdown to indicate the mission impact of the sea state data. If marginal or unfavorable, put in box 21 which parameter is causing the mission impact.

Box 24 – Visibility: Indicate the average visibility along your mission route if the condition is favorable. If the condition is marginal or unfavorable, put the worst expected enroute visibility data.

Box 25 – Precipitation: If there is any forecasted precipitation along your mission route, input the level of mission impact in the dropdown menu, then indicate the type of precipitation and the intensity. Indicate the general location of the precipitation in the last box.

Part 3: Mission Execution Forecast

PART III - Mission Execution Forecast								
26.	27. VALID TIME		28. SFC WIND	29. VSBY/WEA	30. CLOUD LAYERS	31. ALTIMETER	32. MAX TEMP	33. MIN TEMP
DESTINATION	Z	TO	Z			INS	F/C	F/C
ALTN DEST 1	Z	TO	Z			INS	F/C	F/C
ALTN DEST 2	Z	TO	Z			INS	F/C	F/C

Box 26 – Destination / Alternate Destinations: These boxes will contain the location / ICAO of the target area for your mission. If your mission requires multiple locations, use the alternate destination boxes to give mission planners that information.

Box 27 – Valid Time: Input the valid times for the mission in these boxes in Zulu time. These are the times that the mission is expected to operate at its target location.

Box 28 – Surface Winds: These are the winds that are expected at your target location during the valid times. If the valid time covers a significant period, input either the average wind data if nothing is expected to impact operations or put the worst wind conditions expected during the mission window.

Box 29 – Visibility / Weather: Input the forecasted surface visibility and any weather expected at your target location within the valid time. If no weather is expected, input NSW.

Box 30 – Sky Condition: Input the forecasted sky condition for the mission valid times.

Box 31 – Altimeter: Input the lowest altimeter expected for your target location during the mission valid time.

Box 32 – Maximum Temperature: Input the maximum expected temperature is Fahrenheit and Celsius for the target location during the valid times.

Box 33 – Minimum Temperature: Input the minimum expected temperature is Fahrenheit and Celsius for the target location during the valid times.

Part 4 & 5: Comments / Briefing Record

PART IV - COMMENTS/REMARKS		
35. REMARKS		
PART V - BRIEFING RECORD		
36. WX BRIEFED TIME	37. BRIEFERS INITIALS	38. PERSON RECEIVING BRIEF

Box 35 – Remarks: If there is any information you feel the mission planners need to be aware of that was not included in the brief, input that information here.

Box 36 – Weather Briefed Time: The date and time in Zulu you delivered the brief.

Box 37 – Briefer Initials: Input your initials in this block.

Box 38 – Person Receiving Brief Initials: Input either the rank and name of the individual you brief or their initials if that is the information they give you.